Adaptive Jamming Mitigation in Single-Antenna Receivers With Spectral Analysis and Switchable Filtering

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Jamming attacks pose a serious threat to global navigation satellite system single-antenna receivers, and extensive technologies have been developed to mitigate some specific jamming types. In real environments, an adaptive jamming countermeasure should be developed to mitigate the jamming signals due to the lack of knowledge on the existence and types of potential jamming attacks. We propose an integrated method capable of coping with various jamming signals efficiently for single-antenna receivers. The integrated method is implemented by a stand-alone module before the signal processing stage to detect, classify, filter, and monitor varying jamming attacks. In this method, the differential cumulative energy spectrum power metric that describes the spectrum characteristics of the received signals is developed to detect and identify jamming attacks. The adaptive notch filter (ANF) is then improved to filter and dynamically monitor jamming environments through the aid of switchable iteration factors and a check unit. The iteration factors are adjusted for adapting ANF to the identified jamming types. The check unit monitors the state of jamming circumstances and updates the antijamming method by evaluating the ANF performance. The proposed method is tested by semiphysical simulations of common jamming circumstances. It can cope with common jamming attacks with over 21 dB of jamming-to-noise ratio. The C/N_0 improvement is up to 19 dB·Hz, which can be 4.6 dB·Hz higher than the conventional ANF. Moreover, with the proposed method, mitigation performance under varying jamming cases is improved effectively.

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I. INTRODUCTION

Global navigation satellite system (GNSS) has become the backbone infrastructure that supports position, navigation, and timing service [1]. The power of GNSS signals is reduced to around −160 dB when they arrive at a ground receiver, and the GNSS signals are susceptible to lots of interferences [2], [3]. Many interference accidents have been reported in recent years. One example is that the global positioning system (GPS) signals in MOSS Landing Harbor were jammed for more than one month due to the preamplifier jamming of the television antenna, which significantly affected the harbor traffic. During the Hong Kong UAV show in 2018, GPS jamming attacks led to the crashes of 46 drones with a loss of more than $127 000. Jamming attacks not only disturb economic activities but also threaten lives. A NASA report in 2019 outlines an aircraft nearly crashed into a mountain due to a period of GPS jamming [4].

Mitigation of GNSS jamming has been widely investigated. Multiple antenna solution is effective in mitigating the jamming signals [5], [6], [7], [8]; however, external bulky antennas are required in this solution. Extensive efforts have been focused on improving the antijamming capacity for single-antenna receivers since most of the civil receivers are equipped with one antenna. The jamming mitigation for single-antenna receivers can be achieved in two ways, namely, enhancement of GNSS signal processing [9], [10], [11], [12] and excision of the jamming signals [13], [14], [15], [16].

First, the jamming effects can be mitigated with improved signal processing methods, such as vector tracking methods and inertial navigation system (INS) aiding methods. Vector tracking methods use the previous navigation results to couple channels tracking rather than tracking satellites separately, producing a 6 dB antijamming improvement [12], [17]. GNSS/INS integration system can reduce the influence of jamming attacks since INS is a self-contained system [11]. The previous study has shown that the antijamming improvement in deeply coupled GNSS/INS integration is 11 dB for broadband jamming signals [18]. The antijamming improvements of the above signal processing enhancements are less than 11 dB and mainly for broadband noise. Nonnoise jamming [9], [19], [20]; such as continuous wave interference (CWI), sweep CW interference (SCWI), and multiple CW interference (MCWI) is stronger and more destructive than broadband noise, which should be considered in real environments.

Second, jamming excision measures have been developed to cope with nonnoise attacks. These measures usually comprise two steps: 1) jamming detection and identification and 2) jamming mitigation. Jamming detection and identification can be achieved by using postcorrelation or precorrelation algorithms. Postcorrelation algorithms detect the jamming signals considering different features such as the carrier-to-noise ratio (C/N_0) [21], correlator output [22], and positioning errors [23]. However, jamming types cannot be identified since postcorrelation signals have been demodulated. Precorrelation methods monitor raw signals; thus, they are more sensitive to jamming signals. Jamming attacks can be alarmed by monitoring abnormal outputs in automatic gain control [24] and analog-to-digital converter [25]. The detection methods based on monitoring automatic gain control and analog-to-digital converter output are simple but jamming types also cannot be identified. The use of time–frequency analysis delivers characteristics of various jamming signals [20], [26]. To identify jamming types, the power spectral density (PSD) fluctuation [27] is proposed to classify the third harmonic jamming and CWI. However, it is hard to detect other jamming signals such as broadband SCWI [28]. Since time–frequency results of signals show more patterns, different time–frequency analysis techniques, such as short-time Fourier transform [29], Wigner–Villey distribution [30], [31], wavelet packet decomposition [32], and Hilbert–Huang transform [33], have also been developed to classify jamming types. However, their applications are limited by the heavy computational loads.

After identifying the jamming types, mitigation methods are implemented. The robust interference mitigation techniques including time/frequency domain complex signum and frequency/time domain pulse blanking have been used to mitigate chirp attacks in [34]. Most of the robust interference mitigation techniques require data with high quantization bits and complex sampling, and these methods could not be used in the low-cost civil receivers since they commonly use 2-bit quantization and single-phase sampling. A potential countermeasure is to filter the jamming signals. The adaptive notch filters (ANF) can track the jamming frequency and then filter the jamming signals timely [13], [14], [31], [34]. The performance of ANF would degrade under changeable jamming cases since ANF configuration is suitable for specific jamming types. In real circumstances, potential jamming signals may be changed casually. Hence, it is worthwhile to improve ANF performance in mitigating different jamming adaptively.

Despite many previous jamming mitigation studies, little has been conducted on adaptively identifying and countering different jamming cases in an efficient way. As a result, the implementation of jamming mitigation for single-antenna receivers in unknown environments is still a challenge. In this study, an antijamming method implemented by a stand-alone software module before the signal processing stage is developed to detect, identify, and mitigate common jamming attacks in unknown environments. Several works are involved in this study. The differential cumulative energy spectrum power (DCSP) is first proposed to detect and identify jamming types; then, an ANF technology is improved to mitigate and monitor jamming environments through switchable iteration factors and a check unit. The iteration factors are adaptively adjusted based on the DCSP identification results to obtain the satisfied C/N_0. The check unit is designed to monitor the change in jamming cases and update the state of the antijamming method.

The rest of this article is organized as follows. The characteristics of the GNSS signals are analyzed in Section II. Section III presents the framework of the proposed antijamming method in detail. The detection and identification methods based on DCSP and the improved ANF technology are investigated in Sections IV and V, respectively. Test results are presented in Section VI. Finally, conclusions are presented in Section VII.

II. GNSS SIGNAL CHARACTERISTICS UNDER NORMAL AND JAMMING ENVIRONMENTS

This section provides an introduction to the received signals along with details of their frequency characteristics.

A. GNSS Signal Model

The received GNSS signal $r(t)$ can be written as

$$r(t) = \sum_{i=1}^{L} s^i(t) + \eta(t) + \delta t J(t) \tag{1}$$

where L is the number of satellites, $\sum_{i=1}^{L} s^i(t)$ is the sum of satellite signals; $\eta(t)$ is the measurement noise with a variance σ_n^2; $J(t)$ is the jamming signal; and the classification variable δt is introduced to indicate if there are jamming attacks. It means that there are no jamming attacks when the classification variable δt is zero; and it denotes that there are jamming attacks when δt is 1.

Three common jamming signals [9], namely, CWI, MCWI, and SCWI, will be investigated in this study, and they can be modeled as

$$J(t) = \sum_{j=1}^{M} \sqrt{2P_J^j} \sin\left(2\pi t f_J^j(t) + \theta_J^j\right) \tag{2}$$

where P_J^j and θ_J^j are the power and phase delay of the jth jamming component, respectively; $f_J^j(t)$ and M are the frequency and the number of jamming component, respectively. Specifically, CWI is given as $J(t)$ with $M = 1$ and constant frequency f_J; MCWI can be modeled as $J(t)$ with $M > 1$ and corresponding constant f_J^j; and SCWI is given as $J(t)$ with $M = 1$ and $f_J(t)$ varying within $[f_{J,c} - B_J, f_{J,c} + B_J]$, where B_J is bandwidth and $f_{J,c}$ is the center frequency.

B. Characteristics of the Received Signals in Frequency Domain

The key to classifying the jamming signals lies in the fact that the frequency-domain characteristics of jammed signals are different from those of normal GNSS signals. The PSD of the received GNSS signal in (1) can be expressed as

$$R(f) = S(f) + \sigma_n^2 + \delta \cdot J(f) \tag{3}$$

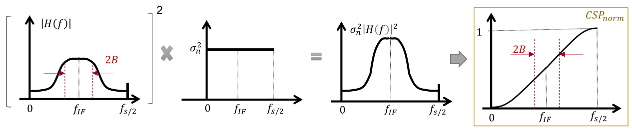


Fig. 1. Illustration of the PSD and $\mathrm{CSP}_{\mathrm{norm}}$ of normal GNSS signals.

where Sf and $J(f)$ are PSDs of the satellite and jamming signals, respectively; σ_n^2 represents the PSD of noise.

Suppose that the transfer function of the front-end conversion process is represented as $H(f)$. The PSD of IF data can be given as

$$\mathrm{PSD}\ (f) = Sf|H(f)|^2 + \sigma_n^2|H(f)|^2 + \delta \cdot J(f)|H(f)|^2. \tag{4}$$

The power variation can be described by the integration of PSD, namely, cumulative spectrum power (CSP)

$$\mathrm{CSP}\left(f\right) = \int_{-\infty}^{f} \mathrm{PSD}\left(f\right) df \ = \int_{0}^{f} \mathrm{PSD}\left(f\right) df \tag{5}$$

where the highest frequency of CSP is its Nyquist frequency $f_s/2$, and thus the total power of IF data is $\mathrm{CSP}(f_s/2)$. In this section, the normalized CSP, i.e., $\mathrm{CSP}_{\mathrm{norm}}(f) = \mathrm{CSP}(f)/\mathrm{CSP}(f_s/2)$, is obtained and analyzed under different scenarios.

1) $\boldsymbol{\delta} = 0$. *Nominal Conditions:* When there are no jamming signals, the PSD of normal signals can be approximated to $\sigma_n^2|H(f)|^2$ since the GNSS signals are much weaker than noise. Therefore, $\mathrm{CSP}_{\mathrm{norm}}(f)$ can be written as

$$\mathrm{CSP}_{\mathrm{norm}}\left(f\right) \approx \frac{\int_0^f \sigma_n^2 \left|H\left(f\right)\right|^2 df}{\int_0^{f_s/2} \sigma_n^2 \left|H\left(f\right)\right|^2 df} = \frac{\int_0^f \left|H\left(f\right)\right|^2 df}{\int_0^{f_s/2} \left|H\left(f\right)\right|^2 df} = \frac{e_H\left(f\right)}{E_H} \tag{6}$$

where $e_H(f)$ and E_H are $e_H(f) = \int_0^f |H(f)|^2 df$ and $E_H = \int_0^{f_s/2} |H(f)|^2 df \ = e_H\ (f_s/2)$, respectively. Fig. 1 shows the PSD and $\mathrm{CSP}_{\mathrm{norm}}$ of normal GNSS signals.

2) $\boldsymbol{\delta} = 1$. *Jammed Conditions:* Jamming signals and noise are much stronger than the GNSS signals, and the PSD of jammed GNSS signals can be simplified as $\sigma_n^2|H(f)|^2 + J(f)|H(f)|^2$. The corresponding $\mathrm{CSP}_{\mathrm{norm}}$ can be written as

$$\mathrm{CSP}_{\mathrm{norm}}\left(f\right) \approx \frac{\int_0^f J\left(f\right)\left|H\left(f\right)\right|^2 + \sigma_n^2 \left|H\left(f\right)\right|^2 df}{\int_0^{f_s/2} \sigma_n^2 \left|H\left(f\right)\right|^2 + J\left(f\right)\left|H\left(f\right)\right|^2 df} = \frac{e_H\left(f\right) + e_J\left(f\right)/\sigma_n^2}{E_H + E_J/\sigma_n^2} \tag{7}$$

where $e_J(f) \ = \int_0^f J(f)|H(f)|^2 df$ is the CSP of the jamming signals; $E_J = e_J\ (f_s/2)$ denotes the total power of the jamming signals from the front end.

Applying PSDs of three jamming signals in (2) into $e_J(f)$, the CSP of different jamming signals can be obtained. The $e_J(f)$ and the resulting $\mathrm{CSP}_{\mathrm{norm}}$ of these jamming signals are shown in Fig. 2. Specifically

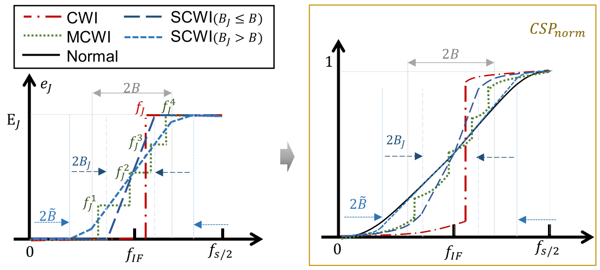


Fig. 2. Illustration of e_J and $\mathrm{CSP}_{\mathrm{norm}}$ of jammed signals.

1) CWI: the CSP e_J is expressed as

$$e_J\left(f\right) = \begin{cases} 0, & f < f_J \\ E_J, & f \geq f_J. \end{cases} \tag{8}$$

The energy of CWI concentrates at f_J, thus the e_J jumps to total power E_J at f_J, which causes the jump at $\mathrm{CSP}_{\mathrm{norm}}(f_J)$.

2) SCWI: the e_J is modeled as

$$e_J\left(f\right) = \begin{cases} 0, \ f < f_{J,c} - \tilde{B} & \\ e_J\left(f\right)|_{f_{J,c}-\tilde{B}}^{f} & \text{other } f \\ E_J, \ f > f_{J,c} + \tilde{B} & \end{cases} \tag{9}$$

where $f_{J,c}$ is simplified to match the GNSS frequency; the e_J of SCWI relates to jamming band $[f_{J,c} - B_J, f_{J,c} + B_J]$ and front-end bandwidth B. $\tilde{B} = B_J$ if $B_J \leq B$: the power of SCWI distributes uniformly in the jamming band. e_J and $\mathrm{CSP}_{\mathrm{norm}}$ increase rapidly in jamming band; $\tilde{B} = \ B + \xi$ if $B_J > B$: ξ denotes the attenuation jamming bandwidth. The out-band jamming signals are attenuated. The power in attenuation band is lower than in-band power. Correspondingly, the increment of e_J and $\mathrm{CSP}_{\mathrm{norm}}$ is slow in attenuation band.

3) MCWI: the e_J is expressed as

$$e_J\left(f\right) = \begin{cases} E_J \frac{i-1}{M}, & f < f_J^i;\ i = 1, 2, \ldots, M \\ E_J, & f \geq f_J^M. \end{cases} \tag{10}$$

The energy of MCWI is distributed at frequency f_J^i of each jamming component. Stepwise increment occurs in the e_J and $\mathrm{CSP}_{\mathrm{norm}}$ of MCWI.

Comparing Figs. 1 and 2, $\mathrm{CSP}_{\mathrm{norm}}$ in different conditions have different patterns. $\mathrm{CSP}_{\mathrm{norm}}$ of normal signals rely on $H(f)$, while $\mathrm{CSP}_{\mathrm{norm}}$ of jammed signals are determined by both $J(f)$ and $H(f)$. This characteristic can be used to distinguish different jamming attacks.

III. Framework of Antijamming Method

Single-antenna receivers have been widely used in many applications, but they can be potentially threatened by different types of jamming attacks. One of the challenges for jamming mitigation in single-antenna receivers is to first know jamming types and then adapt the mitigation tool and its configuration accordingly in a low-cost way. This section presents a lightweight, integrated antijamming method that can detect, identify, and mitigate different jamming

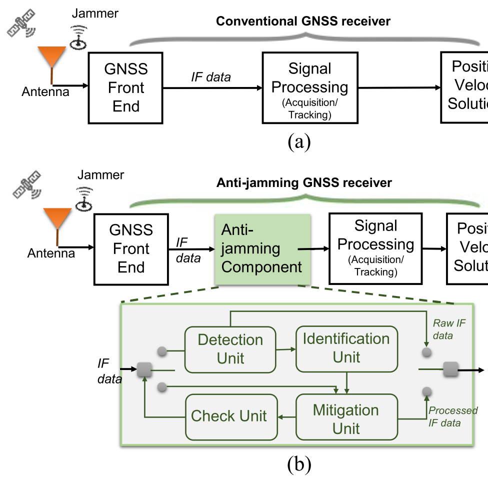


Fig. 3. Framework of (a) conventional receiver and (b) proposed receiver.

attacks efficiently. This stand-alone software module can be incorporated before the signal processing stage to develop antijamming receivers.

Fig. 3(a) shows the block diagram of a conventional GNSS receiver. Fig. 3(b) shows where our proposed algorithm lies in the chain.

The workflow of the antijamming method is as follows. The IF data are obtained from the front end. As the first step, the detection unit finds if there are jamming signals in the IF data. If no jamming attack is detected, raw IF data would be provided to the signal processing unit; otherwise, the types of jamming signals would be classified in the identification unit. Then, according to the identification results, the state of ANF in the mitigation unit is activated to filter the jamming signals. The processed data are finally provided to the signal processing unit. Under this type of jamming signal, ANF tracks and filters the jamming signals continuously, thus the next IF data are input to the mitigation unit directly and the branch of the detection unit is disconnected to avoid redundant computation. More importantly, the check unit monitors the performance of ANF. Once the performance index of ANF is beyond the normal range, the detection unit will be reactivated since the current jamming case changes.

In this mitigation method, the detection and identification units are presented in Section IV. The ANF technique is developed to filter the jamming signals and monitor the state of the jamming environment, which is given in Section V.

IV. DETECTION AND IDENTIFICATION OF JAMMING ATTACKS USING DCSP

The $\mathrm{CSP}_{\mathrm{norm}}$ described in Section II are employed in this section to detect and identify the jamming signals.

A. Jamming Detection

Jamming detection can be carried out based on the fact that $\mathrm{CSP}_{\mathrm{norm}}$ of normal signals in (6) depends on only $H(f)$, while $\mathrm{CSP}_{\mathrm{norm},J}$ of jammed signals in (7) is affected by both $H(f)$ and $J(f)$. In this study, $\mathrm{CSP}_{\mathrm{norm}}$ of the GNSS signals is obtained first, which is denoted as $\mathrm{CSP}_{\mathrm{ref}}$. It will be used as a reference to detect and identify jamming signals. The detection metric $\mathrm{DCSP}_{\mathrm{norm}}(f)$ is obtained by subtracting $\mathrm{CSP}_{\mathrm{ref}}$ from $\mathrm{CSP}_{\mathrm{norm}}$

$$\mathrm{DCSP}_{\mathrm{norm}}\ (f) = \mathrm{CSP}_{\mathrm{norm}}\ (f) - \mathrm{CSP}_{\mathrm{ref}}\ (f)$$
$$= \begin{cases} \sigma\ (f) \\ \frac{[E_H e_J(f) - E_J e_H(f)]}{(\sigma_n^2 E_H{}^2 + E_H E_J)} + \sigma\ (f) = \mu_J\ (f) + \sigma\ (f) \end{cases} \quad (11)$$

where σ represents the noise level 0.

1) *Distribution of* $\mathbf{DCSP_{norm}}$*:* Under nominal conditions, GPS signals are marked by noise in received signals. The PSDs of normal signals demonstrate close alignment with the pattern of the front-end filter, as shown in Fig. 1. The fluctuations in PSD of received signals mainly arise from noise [35], [36]. The noise in normal signals is stable, but its intensity varies across different frequency points due to the process of front-end filtering. Therefore, the PSDs at all frequency points obey normal distributions but have different variations. The variation is high since noise within the passband of the front-end filter remains, while the variation is low since the out-band noise is filtered out.

The normalized CSP represents the integration summation of PSD from 0 Hz to the Nyquist frequency after normalization by the maximum CSP. As per the Central Limit Theorem, the aggregation of independent and identically distributed random variables tends toward a normal distribution [37]. Hence, the normalized CSPs at all frequency points also obey normal distribution. The expectation represents the statistical outcome of $\mathrm{CSP}_{\mathrm{norm}}(f)$ under normal conditions, denoted as $\mathrm{CSP}_{\mathrm{ref}}(f)$.

$\mathrm{DCSP}_{\mathrm{norm}}$ characterizes the difference of the normalized CSP and $\mathrm{CSP}_{\mathrm{ref}}$, effectively centering the expectation of the normalized CSP around zero. Hence, $\mathrm{DCSP}_{\mathrm{norm}}$ still follows normal distributions with zero as expected.

Under jamming conditions, the strength of jamming signals significantly surpasses that of GPS signals. The PSD values at jamming frequencies contain the PSD of noise and jamming signals, resulting in a larger PSD than that of normal signals. As a result, the expectation of the normalized CSP diverges from $\mathrm{CSP}_{\mathrm{ref}}(f)$. In this scenario, the expectation of $\mathrm{DCSP}_{\mathrm{norm}}(f)$ becomes $\mu_J(f)$ rather than zero.

To sum up, under nominal conditions, the metric $\mathrm{DCSP}_{\mathrm{norm}}(f)$ follows a zero-mean normal distribution $N(0, \sigma^2(f))$. Under jamming conditions, the metric $\mathrm{DCSP}_{\mathrm{norm}}$ follows a normal distribution $N(\mu_J(f), \sigma^2(f))$ with a significant bias $\mu_J(f)$.

2) $\boldsymbol{\sigma(f)}$ *of* $\mathbf{DCSP_{norm}}$*:* The noise level $\sigma(f)$ can be obtained from the statistical fluctuation of $\mathrm{DCSP}_{\mathrm{norm}}(f)$ in jamming-free conditions. Fig. 4 shows the standard variations $\sigma(f)$ of $\mathrm{DCSP}_{\mathrm{norm}}$ for jamming-free signals. The front-end bandpass filter causes distinct noise levels at different frequencies. Specifically, within the stopband of filter ($f \in [0,\ f_{\mathrm{IF}} - 2B]$ or $[f_{\mathrm{IF}} + 2B,\ f_s/2]$), σ is low

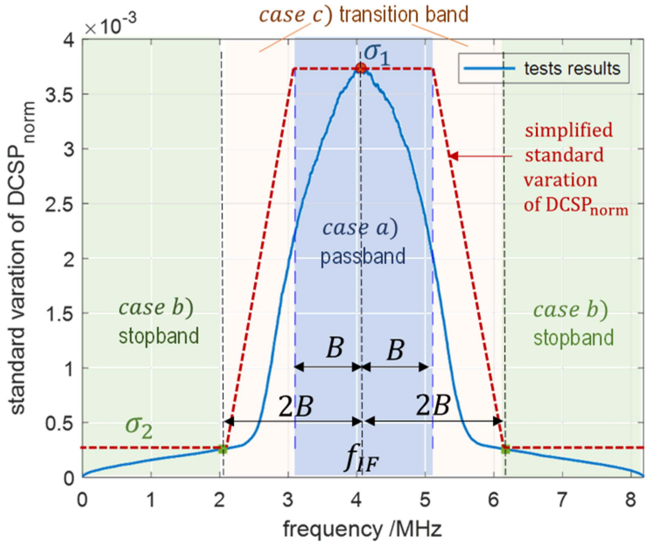


Fig. 4. Standard variations of $\mathrm{DCSP}_{\mathrm{norm}}$ of normal signals.

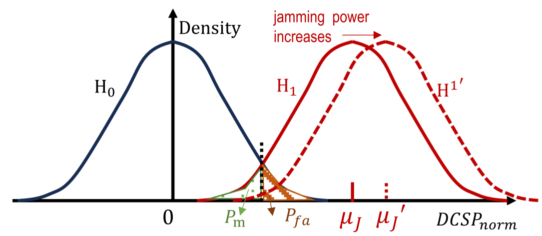


Fig. 5. Density of normal and jamming conditions.

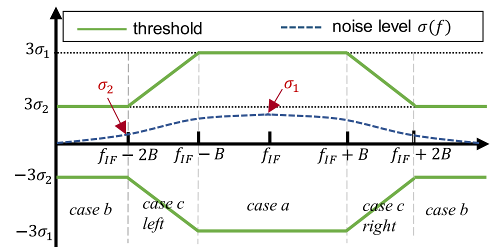


Fig. 6. Diagram of thresholds.

due to out-band noise being filtered. Within the passband ($f \in [f_{\mathrm{IF}}-B,\ f_{\mathrm{IF}}+B]$), noise remains, resulting in a higher σ of $\mathrm{DCSP}_{\mathrm{norm}}(f)$. Within the transition band between the stopband and passband, the variation of $\mathrm{DCSP}_{\mathrm{norm}}(f)$ increases progressively from stopband to passband.

Hence, the standard variation can be simplified into three segments: case a) high level σ_1 in the passband where $\sigma_1 = \max[\sigma(f)]$ and $f \in [f_{\mathrm{IF}}-B,\ f_{\mathrm{IF}}+B]$; case b) low level σ_2 in the stopband where $\sigma_2 = \max[\sigma(f)]$ and $f \in [0,\ f_{\mathrm{IF}}-2B]$or $[f_{\mathrm{IF}}+2B,\ f_s/2]$; case c) a linear increment from σ_2 to σ_1 in the transition band where $f \in (f_{\mathrm{IF}}-2B, f_{\mathrm{IF}}-B)$ or $(f_{\mathrm{IF}}+B, f_{\mathrm{IF}}+2B)$.

The actual $\sigma(f)$ in case c) exhibits a smooth increase, justifying the use of the linear approximations, which have been used to simplify the analysis and design of the transition bands in filters [38]. In addition, σ_1 and σ_2 can encompass the σ levels of the passband and stopband, enabling the linear combination to adequately accounts for the σ level in transition band, as illustrated in Fig. 4. Although the simplified $\sigma(f)$ is slightly larger than the actual results, it can portray the trends of σ. Moreover, it remains much lower than the bias caused by jamming signals, given the notably higher jamming power in comparison to GPS signal power.

3) *Detection Based on* $\mathbf{DCSP_{norm}}$*:* Therefore, the detection of jamming attacks can be cast to a hypothesis test. The hypothesis H_0 represents the nominal condition without jamming attacks, and the alternate hypothesis H_1 represents a jamming condition. Given the challenge of accurately estimating the prior probability of a jamming attack, we adopt the Neyman–Pearson paradigm [39] to perform the hypothesis test.

Fig. 5 depicts the distribution schematic of $\mathrm{DCSP}_{\mathrm{norm}}$ under hypotheses H_0 and H_1, where P_{fa} and P_m denote the false alert probability and missed detection probability, respectively. Typically, our objective is to minimize P_m while constraining that P_{fa} remains within a satisfied level

$$\min_{|\lambda_{\mathrm{th}}(f)|} P\left(\mathrm{DCSP}_{\mathrm{norm}}(f) \geq \left|\lambda_{\mathrm{th}}(f)\right| \middle| \mathrm{H}_1\right)$$
$$\text{s.t. } P\left(\mathrm{DCSP}_{\mathrm{norm}}(f) < \left|\lambda_{\mathrm{th}}(f)\right| \middle| \mathrm{H}_0\right) \leq P_{fa} \qquad (12)$$

where λ_{th} denotes the threshold. If $\mathrm{DCSP}_{\mathrm{norm}}$ exceeds the threshold, the jamming signals are alarmed.

The relationship between jamming power and the parameter μ_J is crucial for threshold determination. As jamming power increases, μ_J correspondingly increases, leading to a decrease in P_m, as illustrated in Fig. 5. Hence, the threshold is mainly determined by P_{fa}. In this study, an example threshold under $P_{fa} \leq 0.3\%$ is derived as three times of the standard deviation (σ) for a normal distribution.

Fig. 6 shows the threshold $\lambda_{\mathrm{th}}(f)$ diagram with three times of $\sigma(f)$, and its expression is

$$\left|\lambda_{\mathrm{th}}(f)\right| = \begin{cases} 3\sigma_1, & case\ a \\ 3\sigma_2, & case\ b \\ 3\left(\frac{\sigma_1-\sigma_2}{B}f + \sigma_2 - \frac{\sigma_1-\sigma_2}{B}\left(f_{\mathrm{IF}}-2B\right)\right), & case\ c, left \\ 3\left(\frac{\sigma_2-\sigma_1}{B}f + \sigma_2 - \frac{\sigma_2-\sigma_1}{B}\left(f_{\mathrm{IF}}+2B\right)\right), & case\ c,\ right. \end{cases} \qquad (13)$$

Notably, the acceptable false alert probability should be considered seriously according to different applications and industries. Aviation, for instance, demands extremely high reliability and stringent safety standards, which directly affects the tolerance for false alerts or errors. Therefore, systems used in these fields require remarkably low false alert probabilities to ensure operational safety and reliability.

B. Jamming Identification

Identifying the jamming types helps the configuration of the jamming mitigation approach. The characteristics of the detection metric $\mathrm{DCSP}_{\mathrm{norm}}$ are further exploited to identify the jamming signals: CWI, SCWI, and MCWI.

Substituting the $\mathrm{DCSP}_{\mathrm{norm}}$ into (11) with e_J of three jamming signals in (8)–(10), the expectation of $\mathrm{DCSP}_{\mathrm{norm}}$ under different jamming scenarios can be obtained.

1) CWI: $\mathrm{DCSP}_{\mathrm{norm}}$ is derived as

$$E\left\{\mathrm{DCSP}_{\mathrm{norm}}\right\} = \begin{cases} \frac{-E_J e_H(f)}{\left(\sigma_n^2 {E_H}^2 + E_H E_J\right)} = -\Upsilon\left(f\right), \; f < f_J \\ \frac{E_J\left[E_H - e_H(f)\right]}{\left(\sigma_n^2 {E_H}^2 + E_H E_J\right)} = \Upsilon\left(\frac{f_s}{2}\right) - \Upsilon\left(f\right), f \geq f_J. \end{cases} \tag{14}$$

From (14), the function $\Upsilon(f)$ is solely related to $e_H(f)$ and consistently rises with increasing frequency. Its maximum value occurs at $\Upsilon(f_s/2)$. As the frequency varies from 0 to $f_s/2$, $\mathrm{DCSP}_{\mathrm{norm}}$ of CWI diminishes to the minimum value of $-\Upsilon(f_J)$ at f_J, and then decreases from the maximum value $\Upsilon(f_s/2) - \Upsilon(f_J)$ at f_J to 0.

2) SCWI: $\mathrm{DCSP}_{\mathrm{norm},J}$ is expressed as

$$E\left\{\mathrm{DCSP}_{\mathrm{norm}}\right\} = \begin{cases} \frac{-E_J e_H(f)}{\left(\sigma_n^2 {E_H}^2 + E_H E_J\right)} = -\Upsilon\left(f\right), \; f < f_{J,c} - \tilde{B} \\ \frac{E_J\left(E_H - e_H(f)\right)}{\left(\sigma_n^2 {E_H}^2 + E_H E_J\right)} = \Upsilon\left(\frac{f_s}{2}\right) - \Upsilon\left(f\right), \; f > f_{J,c} + \tilde{B} \\ \frac{E_H E_J\left[e_J(f)/E_J - e_H(f)/E_H\right]}{\left(\sigma_n^2 {E_H}^2 + E_H E_J\right)} = \Upsilon\left(\frac{f_s}{2}\right)\left[\frac{e_J(f)}{E_J} - \frac{e_H(f)}{E_H}\right], \\ \qquad \text{other } f. \end{cases} \tag{15}$$

From (15), $\mathrm{DCSP}_{\mathrm{norm}}$ of SCWI decreases outside the range $[f_{J,c} - \tilde{B},\ f_{J,c} + \tilde{B}]$, while the patterns of $\mathrm{DCSP}_{\mathrm{norm},J}$ in $[f_{J,c} - \tilde{B},\ f_{J,c} + \tilde{B}]$ depends on B_J and B. Referring to Fig. 2, when $B_J \leq B$, $\tilde{B} = B_J$ and the quantity $[e_J(f)/E_J - e_H(f)/E_H]$ increases from $f_{J,c} - \tilde{B}$ to $f_{J,c} + \tilde{B}$. Consequently, $\mathrm{DCSP}_{\mathrm{norm}}$ of SCWI ($B_J \leq B$) increases from the minimum to the maximum. However, if $B_J > B$, then $\tilde{B} < B$ and $[e_J(f)/E_J - e_H(f)/E_H]$ fluctuate within $[f_{J,c} - \tilde{B}, f_{J,c} + \tilde{B}]$. Consequently, $\mathrm{DCSP}_{\mathrm{norm}}$ of SCWI ($B_J > B$) also exhibits fluctuations.

3) MCWI: $\mathrm{DCSP}_{\mathrm{norm}}$ is obtained as

$$E\left\{\mathrm{DCSP}_{\mathrm{norm}}\right\} = \begin{cases} \frac{E_J\left[E_H \frac{i-1}{M} - e_H(f)\right]}{\left(\sigma_n^2 {E_H}^2 + E_H E_J\right)} = \frac{i-1}{M}\Upsilon\left(\frac{f_s}{2}\right) - \Upsilon\left(f\right), \\ \qquad f < f_J^i, \; i = 1, 2, \ldots, M \\ \frac{E_J\left(E_H - e_H(f)\right)}{\left(\sigma_n^2 {E_H}^2 + E_H E_J\right)} = \Upsilon\left(\frac{f_s}{2}\right) - \Upsilon\left(f\right), \; f \geq f_J^M. \end{cases} \tag{16}$$

From (16), as the frequency increases from 0 to f_J^1, $\mathrm{DCSP}_{\mathrm{norm}}$ of MCWI decreases from 0 to the minimum $-\Upsilon(f_J^1)$. Subsequently, it jumps to a local maximum $\frac{1}{M}\Upsilon(f_s/2) - \Upsilon(f_J^1)$. With further frequency increases from f_J^{i-1} to f_J^i ($i > 1$), $\mathrm{DCSP}_{\mathrm{norm}}$ decreases from a previous local maximum $\frac{i-1}{M}\Upsilon(f_s/2) - \Upsilon(f_J^{i-1})$ to a local minimum $\frac{i-1}{M}\Upsilon(f_s/2) - \Upsilon(f_J^i)$, then jumps to a local maximum $\frac{i}{M}\Upsilon(f_s/2) - \Upsilon(f_J^i)$. Finally, as the frequency progresses

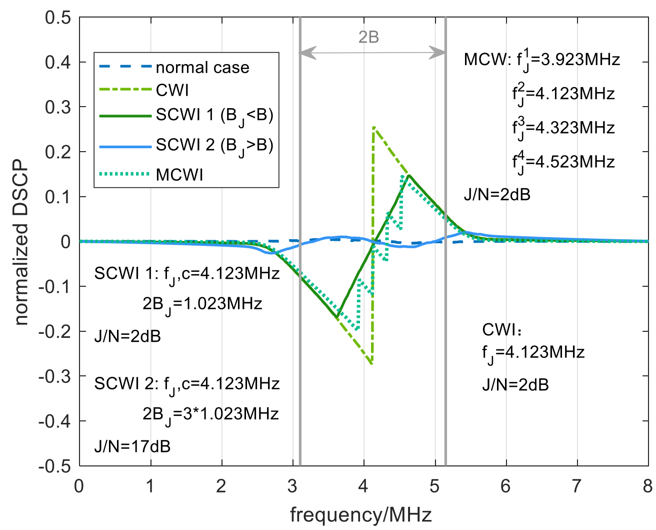


Fig. 7. $\mathrm{DCSP}_{\mathrm{norm}}$ under the various conditions.

from f_J^M to $f_s/2$, $\mathrm{DCSP}_{\mathrm{norm}}$ of MCWI decreases from the maximum $\Upsilon(f_s/2) - \Upsilon(f_J^M)$ to 0.

To validate the theoretical analysis of $\mathrm{DCSP}_{\mathrm{norm}}$, GPS data under various jamming signals are simulated to calculate their $\mathrm{DCSP}_{\mathrm{norm}}$, according to (11). The mean of 2000 $\mathrm{CSP}_{\mathrm{norm}}$ of real jamming-free data is used as the reference $\mathrm{CSP}_{\mathrm{ref}}$. The variations of $\mathrm{DCSP}_{\mathrm{norm}}$ across the frequency range $[0,\ f_s/2]$ are plotted in Fig. 7. The parameters of jamming signals are provided in the figure and the sampling frequency is $f_s = 16.367667$ MHz. A detailed explanation of the simulation process is given in Section VI-A.

The test results of $\mathrm{DCSP}_{\mathrm{norm}}$ in Fig. 7 align consistently with the analysis outlined in (13)–(15). Specifically, $\mathrm{DCSP}_{\mathrm{norm}}$ of CWI experiences a jump from the minimum to the maximum at f_J. $\mathrm{DCSP}_{\mathrm{norm}}$ of SCWI ($B_J \leq B$) increases from the minimum at $f_{J,c} - B_J$ to the maximum at $f_{J,c} + B_J$, while that of SCWI ($B_J > B$) fluctuates within $[f_{J,c} - \tilde{B}, f_{J,c} + \tilde{B}]$. $\mathrm{DCSP}_{\mathrm{norm}}$ of MCWI decreases to the minimum at f_J^1 and similarly drops from the maximum after f_J^4. At $f = f_J^2$ *or* f_J^3, $\mathrm{DCSP}_{\mathrm{norm},J}$ jumps to local maxima and then declines to local minima.

Since different jamming signals have different patterns of $\mathrm{DCSP}_{\mathrm{norm}}$, the jamming type can be identified through analyzing the features of $\mathrm{DCSP}_{\mathrm{norm}}$ in the frequency domain. After detecting jamming signals using the method outlined in Section IV-A, the identification of the jamming type can be realized according to the following criteria: where $f_{\max}$ and $f_{\min}$ denote the frequencies corresponding to the maximum $\mathrm{DCSP}_{\mathrm{norm}}$ and the minimum $\mathrm{DCSP}_{\mathrm{norm}}$; $f_{\mathrm{loc\_max}}^i$ and $f_{\mathrm{loc\_min}}^i$ represent the frequencies of the local maximum and the local minimum of $\mathrm{DCSP}_{\mathrm{norm}}$ if corresponding spectrum exists; ∇ signifies the frequency interval of $\mathrm{DCSP}_{\mathrm{norm}}$.

CWI,	if	$f_{max} - f_{min} = \nabla$;
MCWI,	if and	$f_{max} - f_{min} > \nabla$ $f^i_{loc\_max} - f^i_{loc\_min} = \nabla$;
SCWI, $(B_J \leq B)$	if and	$f_{max} - f_{min} > \nabla$ $D\dot{C}SP_{norm}(f_{min}{:}f_{max}) > 0$;
SCWI, $(B_J > B)$	if and	$f_{max} - f_{min} > \nabla$ $f_{max} - f_{min} > 2B$.

Calculation of ∇*:* The frequency interval ∇ of $\mathrm{DCSP}_{\mathrm{norm}}$ aligns with the frequency interval of PSD. Welch's method [40], utilized for PSD estimation, determines the interval as $\nabla = f_s/\mathrm{NFFT}$, where NFFT represents the FFT points.

Identifying the local maxima and minima: The $f_{\min}$ and $f_{\max}$ are set as the frequencies of the first local minimum $f^1_{\mathrm{loc\_min}}$ and the last local maximum $f^{\mathrm{end}}_{\mathrm{loc\_max}}$, respectively. Then, the first derivative of $\mathrm{DCSP}_{\mathrm{norm}}$ in the frequency range $[f_{\min}, f_{\max}]$ is calculated to assess the changes in slope direction, represented as $\mathrm{D\dot{C}SP}_{\mathrm{norm}}(f_{\min} : f_{\max})$. At a local maximum, the derivative transitions from a positive to a negative value. Hence, if $\mathrm{D\dot{C}SP}_{\mathrm{norm}}(k-1) > 0$ and $\mathrm{D\dot{C}SP}_{\mathrm{norm}}(k) < 0$, a local maximum is identified as $\mathrm{DCSP}_{\mathrm{norm}}(k)$. Conversely, for a local minimum, the derivative changes from negative to positive values. Thus, if $\mathrm{D\dot{C}SP}_{\mathrm{norm}}(i-1) < 0$ and $\mathrm{D\dot{C}SP}_{\mathrm{norm}}(i) > 0$, the local minimum is searched as $\mathrm{DCSP}_{\mathrm{norm}}(k)$. If local extrema appear in pairs, the frequencies of local minimum and the subsequent local maximum are marked as $f^i_{\mathrm{loc\_min}}$ and $f^i_{\mathrm{loc\_max}}$, where $i = 1, 2, \ldots, M$ represents the ith local extrema group. If $f^i_{\mathrm{loc\_max}} - f^i_{\mathrm{loc\_min}} = \nabla$, then the jamming type is identified as MCWI, and the number of jamming components in MCWI is M.

V. JAMMING MITIGATION AND MONITOR METHODOLOGIES BASED ON IMPROVED ANF

After knowing the jamming types, the corresponding mitigation tools should be performed to reduce their effects on the GNSS receiver. ANF is used in this study to track and filter the different jamming signals. The transfer function of an ANF is presented as [13], [41]

$$H_{\mathrm{ANF}}(z) = \frac{N(z)}{D(z)} = N(z) H_p z = \frac{1 - 2\alpha z^{-1} + z^{-2}}{1 - \alpha(1+k) z^{-1} + k z^{-2}} \tag{17}$$

where $k \in (0, 1)$ determines the notch bandwidth. Previous studies [14], [15] showed that ANF with a narrow bandwidth could reduce the loss of useful signals. In this study, $k = 0.98$ is used, which is adopted in previous studies [15]. Notch frequency f_{stop} depends on variable $\alpha = \cos(2\pi f_{\mathrm{stop}}/f_s)$. The adaptive tracking and filtering of the jamming signals in ANF are achieved by the update of α, which are detailed in previous studies [13], [41]

$$\alpha[n] = \gamma\alpha[n-1] + (1-\gamma)\, c[n]/d[n]$$

$$\begin{cases} c[n] = \lambda c[n-1] + (1-\lambda)\, x[n-1]\,(x[n] + x[n-2]) \\ d[n] = \lambda d[n-1] + (1-\lambda)\, 2x[n-1]^2 \end{cases} \tag{18}$$

where $x[n] = r[n] H_p z = r[n]/D(z)$ is the output of all-pole sections and $r[n]$ represents input signals; the forgetting factor λ and smoothing factor γ reflect iteration step of notch frequency. The variable $d[n]$ indicates the energy of $x[n]$.

For CWI and SCWI, a single ANF will be activated since CWI or SCWI signals are typically single-frequency jamming signals. In case of identifying MCWI with M jamming components, M of ANFs will be activated and configured in a cascaded manner. ANFs are able to filter the signal components with high energy [41], and by cascading them, jamming components in MCWI can be filtered sequentially. The cascaded structure avoids multiple filters simultaneously tracking the same jamming components. Research in [42] also demonstrates the effectiveness of the cascaded ANF in tracking multiple jamming components.

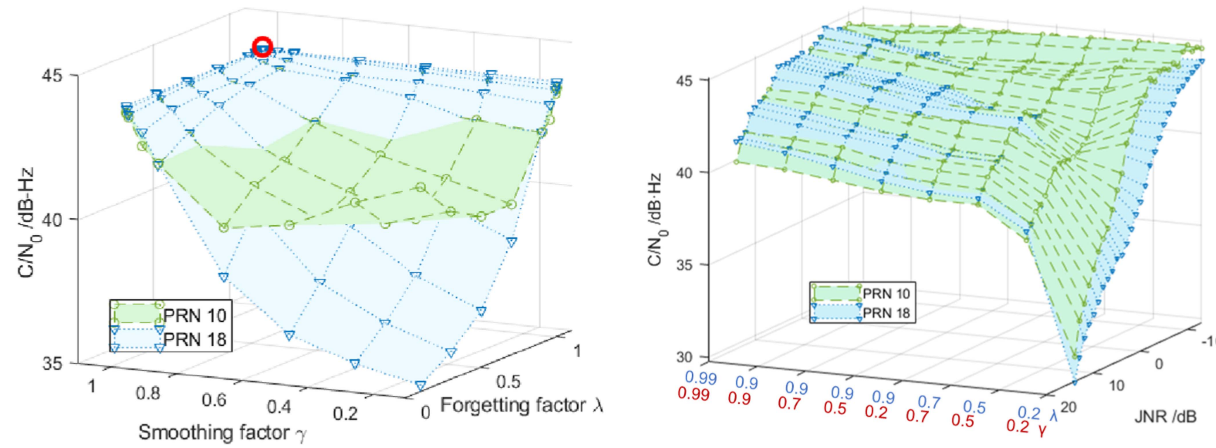
However, the conventional ANF generally uses fixed iteration factors for specific jamming cases and its performance would degrade when jamming attacks are changed. To cope with different and varying jamming signals, this study improves the ANF through: 1) switchable iteration factors (λ, γ) for different jamming attacks by lookup table and 2) monitoring the jamming variation by variable $d[n]$.

A. Improved ANF With Switchable Iteration Factors

When the jamming signal is filtered by ANF, the C/N_0 of the useful GNSS signals can be improved. The C/N_0 level of the processed data depends on the effectiveness of the filter. Less errors in estimating the notch frequency related to jamming frequency can achieve better filter performance of the jamming signals.

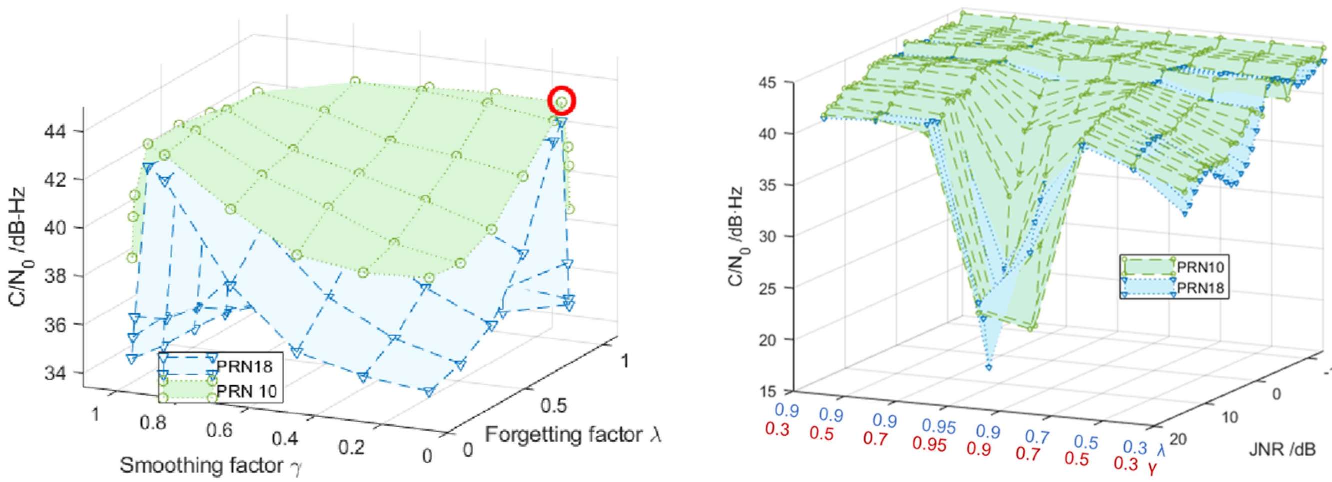
The iteration factors λ and γ affect the estimation of the notch frequency. When both λ and γ are high, the notch frequency is heavily influenced by previous results. If λ and γ are small, random errors of estimated jamming frequency could not be smoothed. Hence, the iteration factors λ and γ should be configured according to the variation frequency characteristics of jamming attacks. Figs. 8 and 9 show the C/N_0 as a function of factors λ and γ under different jamming attacks with various JNRs. Meanwhile, Tables I and II present the specific C/N_0 values corresponding to different iteration factors. The λ and γ parameters for specific jamming signals are expected to provide satisfactory C/N_0 values across different JNR conditions.

As shown in Fig. 8(a), as CWI frequency is stable, higher weights of previous estimations can bring a more stable and smoother estimation results. When both γ and λ are over 0.99, C/N_0 reaches to the maximum. Moreover, as the influence of γ and λ on C/N_0 exhibits symmetrical behavior, eight typical iteration factor combinations are represented to assess the C/N_0 performance under different JNR levels, as depicted in Fig. 8(b). Notably, given the similar C/N_0 patterns across different PRN channels,



(a) (b)

Fig. 8. Influence of iteration factors on C/N_0 under CWI attacks. (a) JNR = 2 dB. (b) Increasing JNR.



(a) (b)

Fig. 9. Influence of iteration factors on C/N_0 under SCWI attacks. (a) JNR = 2 dB. (b) Increasing JNR.

TABLE I

C/N_0 [dB·Hz] Comparisons of Different Iteration Factors for GNSS Under CWI Attacks (PRN 18)

iteration factors		**JNR [dB]**								
λ	**γ**	**-13**	**-10**	**-7**	**-4**	**-1**	**2**	**5**	**8**	**11**
0.2	**0.2**	43.9	43.3	42.2	40.7	38.5	36.5	35.3	34.1	32.4
0.5	**0.5**	43.9	43.3	42.2	40.9	39.5	39.3	39.9	39.2	38.5
0.7	**0.7**	43.8	43.3	42.4	42.2	41.9	42.1	42.4	41.0	40.3
0.9	**0.2**	43.9	43.3	42.5	42.5	42.0	42.1	42.4	41.1	40.3
0.9	**0.3**	43.8	43.4	42.9	43.1	42.4	42.8	42.6	41.6	41.0
0.9	**0.7**	43.8	43.6	43.3	43.7	42.8	43.3	43.2	41.9	41.3
0.9	**0.9**	43.7	43.9	43.7	44.2	43.2	43.4	43.6	42.2	41.7
0.99	**0.99**	43.7	44.2	44.1	44.5	43.7	43.9	43.9	42.6	42.1

Table I further provides specific C/N_0 values of PRN 18 as an example. Under low JNR conditions, such as -13 dB, the degradation of C/N_0 caused by CWI is slight [43], resulting in similar C/N_0 whether CWI is filtered or not. Consequently, the influence of iteration factors on C/N_0 is not significant, with values ranging from 43.7 to 43.9 dB·Hz. As JNR increases, the combination of $\gamma = 0.99$ and $\lambda = 0.99$ consistently exhibits satisfactory C/N_0 performance. Since the frequency variation of MCWI is also slight, the optimal factors for CWI filtering are also suitable for that of MCWI. Therefore, the optimal λ and γ for CWI and MCWI are 0.99.

TABLE II
C/N_0 [dB·Hz] Comparisons of Different Iteration Factors for GNSS Under SCWI Attacks (PRN 18)

iteration factors		**JNR [dB]**								
λ	**γ**	**-13**	**-10**	**-7**	**-4**	**-1**	**2**	**5**	**8**	**11**
0.3	**0.3**	43.2	42.3	42.4	43.5	41.1	36.4	35.2	36.7	36.8
0.5	**0.5**	43.2	42.4	42.4	43.5	41.4	38.9	38.8	39.7	39.9
0.7	**0.7**	43.2	42.4	42.7	43.4	42.8	42.3	41.3	41.1	40.6
0.9	**0.9**	43.2	42.6	42.2	42.4	42.1	39.7	36.6	35.3	35.2
0.95	**0.95**	43.1	42.3	42.4	42.8	41.8	37.0	32.4	27.9	20.0
0.9	**0.7**	43.2	42.9	43.2	43.5	43.3	42.8	40.8	39.4	39.5
0.9	**0.5**	43.2	42.7	43.0	43.4	43.2	42.8	41.4	40.5	40.4
0.9	**0.3**	43.2	42.6	42.9	43.4	43.1	42.6	41.4	41.0	40.5

The C/N_0 comparisons of different iteration factors under SCWI attacks are shown in Fig. 9 and Table II. In Fig. 9(a), there is a tradeoff of λ and γ to track the varying frequency of SCWI with 2 dB of JNR. Specifically, the λ of 0.9 and γ of 0.3 are regulated to achieve the optimal C/N_0. Similar to CWI attacks, the C/N_0 results of λ and γ are symmetrical. Hence, Fig. 9(b) illustrates the C/N_0 performance of typical iteration factors under SCWI attacks with increasing JNRs, while Table II provides the specific C/N_0 values. When JNR is below -1 dB, the C/N_0 degradation caused by SCWI signals is slight, as proved in [43]. The efficacy of filtering SCWI signals, which is influenced by iteration parameters, has little impact on C/N_0. Therefore, under low JNR conditions, C/N_0 remains similar across different iteration factor combinations, ranging from 42 to 43.5 dB·Hz. As JNR increases from 1 dB to 11 dB, three combinations—λ $= 0.9$ and $\gamma = 0.3$; λ $= 0.9$ and $\gamma = 0.5$; λ $= 0.7$ and $\gamma = 0.7$—consistently exhibit high C/N_0 values. Therefore, these three factor combinations can be used for SCWI attacks. In this study, considering that under 2–11 dB of JNR conditions, the C/N_0 performance of λ $= 0.9$ and $\gamma = 0.3$ surpasses that of the other two factor combinations. Hence, the iteration factors are set to λ at 0.9 and γ at 0.3 for SCWI attacks.

B. Jamming State Monitor by a Check Unit

As the type of jamming signals may be changed in real conditions, the state of jamming attacks should be monitored to configure ANF accordingly. A check unit in Fig. 3(b) is developed in this study to monitor jamming circumstance by evaluating the performance of ANF.

The performance of ANF can be reflected by its filtered data. From (17) and (18), the filtered data are the output of all-pole sections H_pz and the variable $d[n]$ indicates the energy of the filtered data. Fig. 10 shows the frequency response of H_pz with three different parameters α in (17). The Nyquist frequency $f_s/2$ serves as the unit frequency. Any frequency parameter f is normalized by the Nyquist frequency, denoted as $(2f/f_s)$. The normalized frequency is multiplied by π to convert to angular frequency around the unit circle. The frequency responses demonstrate that H_pz acts as an extremely narrow bandpass filter with pass

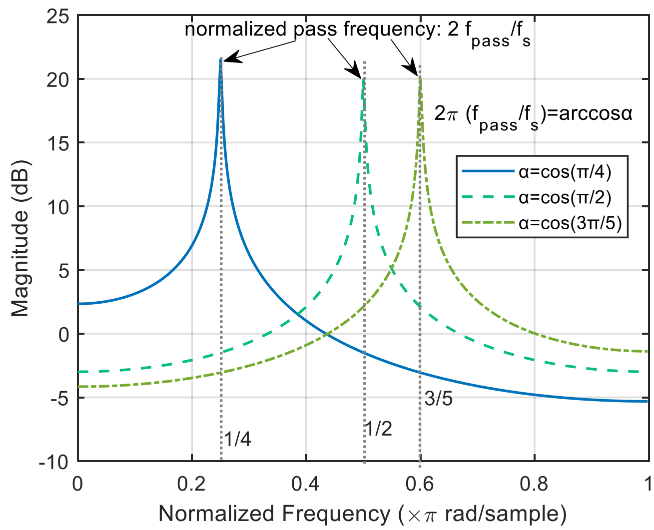


Fig. 10. Frequency response of H_pz.

frequency f_{pass}. The normalized frequency at peak represents the normalized pass frequency, i.e., $2\pi f_{\text{pass}}/f_s$, which equals $\arccos(\alpha)$. Previous analysis of (17) reveals that the parameter α is also equivalent to $\cos(2\pi f_{\text{stop}}/f_s)$. Hence, the pass frequency of H_pz aligns with f_{stop} of ANF.

Therefore, the variable $d[n]$ is used in the check unit to assess the filtered data and to measure states of jamming cases. If the jamming signals remain unchanged, ANF with specific configurations can continuously filter the jamming signals; $d[n]$ is high due to large jamming power. When the jamming type changes, the ANF fails to track the jamming signals since its configuration is not suitable for the new jamming attack, and $d[*]$ drops to the energy level of normal signals.

If the jamming state remains unchanged and the jamming signal can be filtered, $d[n]$ can be represented as

$$\begin{aligned} d[n] &= 2E\left[(r[n]/D(z))^2\right] = 2\times r f_{\text{stop}} \left|H_p\left(f_{\text{stop}}\right)\right|^2 \\ &\approx 2\sigma_n^2 \left|H_p\left(f_{\text{stop}}\right)\right|^2 + J\left(f_{\text{stop}}\right)\left|H_p\left(f_{\text{stop}}\right)\right|^2 \end{aligned} \quad (19)$$

where $S(f_{\text{stop}})|H(f_{\text{stop}})|^2$is ignored since GNSS power is very low. $d[*]$ follows normal distribution $N(\mu_{d,J}, \sigma_{d,J}^2)$.

The $\mu_{d,J} = 2\sigma_n^2|H_p(f_{\text{stop}})|^2 + J(f_{\text{stop}})|H_p(f_{\text{stop}})|^2$ is the expectation and the variation $\sigma_{d,J}^2$ refers to the fluctuation of noises and the jamming signals.

When the jamming type is changed and the jamming signal is failed to be filtered, $d[*]$ can be written as

$$d\left[n\right] = 2 \times r f_{\text{stop}}\left|H_p\left(f_{\text{stop}}\right)\right|^2 \approx 2\sigma_n^2\left|H_p\left(f_{\text{stop}}\right)\right|^2. \quad (20)$$

In this circumstance, $d[*]$ follows normal distribution $N(\mu_{d,n}, \sigma_{d,n}^2)$. The expectation $\mu_{d,n}$ is $2\sigma_n^2|H_p(f_{\text{stop}})|^2$ and the variation $\sigma_{d,n}^2$ refers to the fluctuation of noise.

Therefore, the ANF performance monitored by the check unit can also be regulated to a hypothesis test: the hypothesis H_0, jamming attack changes; the hypothesis H_1, jamming attack is unchanged.

$$\begin{cases} H_0 : d\left[n\right] \sim \mathbb{N}\left(\mu_{d,n}, \sigma_{d,n}^2\right) \\ H_1 : d\left[n\right] \sim \mathbb{N}\left(\mu_{d,J}, \sigma_{d,J}^2\right). \end{cases} \quad (21)$$

Similar to the jamming detection problem in Section IV, the priori probabilities of H_1 are hard to obtain and the hypothesis test is performed based on Neyman–Pearson tests [39]. The P_{fa} is constrained and by minimizing P_m, hence, the threshold can be calculated by

$$P_{fa} = \int_{\lambda_{d,\text{th}}}^{\infty} p_{H_0}\left(d\left[n\right]|H_0\right) \, \mathrm{d}d\left[n\right] \quad (22)$$

where p_{H_0} is the density function of the hypothesis H_0. Given $P_{fa} = 0.3\%$ as an example, the threshold $\lambda_{d,\text{th}}$ is derived as $\mu_{d,n} + 3\sigma_{d,n}$. From (19) and (20), the expectation $\mu_{d,n}$ under hypothesis H_0 is constant and only related to noise, while $\mu_{d,J}$ under the hypothesis H_1 is determined by both noise and jamming power. Hence, $\mu_{d,J} \gg \mu_{d,n}$ and the P_m decreases with the increment of jamming power. It is important to note that the selection of P_{fa} also depends on the application and it will be much lower in stricter field, such as aviation. The $\mu_{d,n}$ and $\sigma_{d,n}^2$ can be obtained from the statistical value in jamming-free case. If variable $d[n]$ is less than $\lambda_{d,\text{th}}$, the filtered signals are not the jamming signals and jamming cases may change. The check unit will activate the detection unit to reassess the received signals.

VI. SIMULATIONS

In this section, the performances of the proposed anti-jamming method are evaluated.

A. Test Description

To assess the performance of the proposed method, we perform semiphysical simulations of common jamming environments with the use of the real GNSS data and a GNSS jamming simulator [44], as shown in Fig. 11. The parameters of jamming attacks are given in Table III.

The real jamming-free GPS data are collected by Newstar 210M front-end. Its sampling frequency is 16.367667 MHz and IF is 4.123968 MHz. The bandwidth of the front-end filter is $B = 1.023$ MHz and 2-bit quantization bits are applied. The jammed signals are generated by GNSS jamming simulator. The code, carrier, and navigation data of real GPS are extracted to generate pure GPS signals,

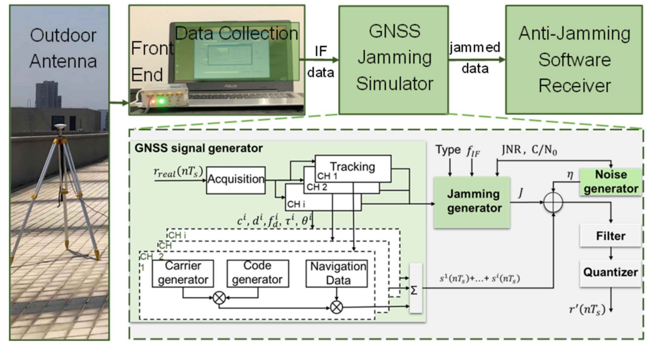


Fig. 11. Diagram of GNSS jamming simulation.

TABLE III
Settings of GNSS Jamming Attacks

Jamming No	CWI f_J [MHz]	MCWI* M	Step[MHz]	f_{init}[MHz]	SCWI* B_J[MHz]
1	f_{IF}	3	0.2	$f_{IF} - 0.1$	$B/2$
2	$f_{IF} + 0.1$	4	0.1	$f_{IF} - 0.1$	B
3	$f_{IF} - 0.1$	5	0.15	$f_{IF} - 0.3$	$3B/2$

(MCWI*: f_{init} denotes the minimum frequency of MCWI;
SCWI*: $f_{J,c} = f_{IF}$, $T_J = 1$ ms)

and then the pure GPS data are modulated with noise and the jamming signals. After the filter and quantization, the jammed GNSS signals are simulated.

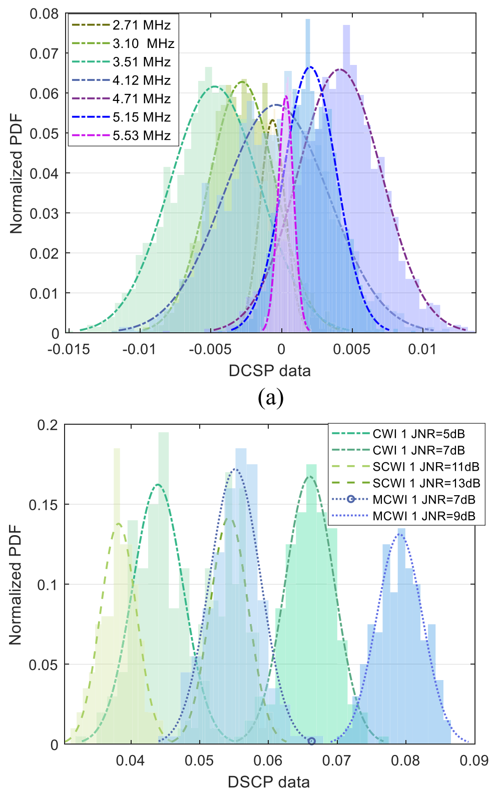
B. Performance of Jamming Detection and Identification

The DCSP$_{\text{norm}}$ distributions of normal signals at several frequency are presented in Fig. 12(a). The mean of 2000 CSP$_{\text{norm}}$ of jamming-free conditions is used as the reference CSP$_{\text{ref}}$. All DCSP$_{\text{norm}}$s follow the normal distributions with expectations in $[-0.005, 0.005]$ and with variations below 0.00352. Slight derivations in the expectations are contributed by noise and hardware jitter. In addition, since out-band noise is filtered by the front-end filter, the variation of out-band DCSP$_{\text{norm}}$s (e.g., 2.71 and 5.53 MHz) is less than that of in-band DCSP$_{\text{norm}}$s (e.g., 3.100–5.149 MHz).

As analyzed in Section IV, DCSP$_{\text{norm}}$s under the jamming conditions tends to peak at the largest jamming frequency. The DCSP$_{\text{norm}}$ distributions at the largest frequency of CWI, MCWI, and SCWI are shown in Fig. 12(b). The DCSP$_{\text{norm}}$s of the jammed signals also follow normal distributions with similar variations, with a much larger bias than that of DCSP$_{\text{norm}}$s in normal signals. As jamming power grows, the expectation of DCSP$_{\text{norm}}$ increases correspondingly.

The threshold is determined using the method in Section IV. Fig. 13 illustrates the relationship of various DCSP$_{\text{norm}}$ with designed threshold lines. DCSP$_{\text{norm}}$ in normal cases are within the threshold envelope, while DCSP$_{\text{norm}}$ in jamming cases exceed the threshold. The jamming signals with higher JNRs lead to more serious excess.

The jamming attacks are detected by comparing DCSP$_{\text{norm}}$ with the designed threshold. In the test of normal signals, no alert was raised. The detection rates of



(a)

(b)

Fig. 12. Histograms of $\mathrm{DCSP}_{\mathrm{norm}}$ under different conditions. (a) Normal case. (b) Jamming case.

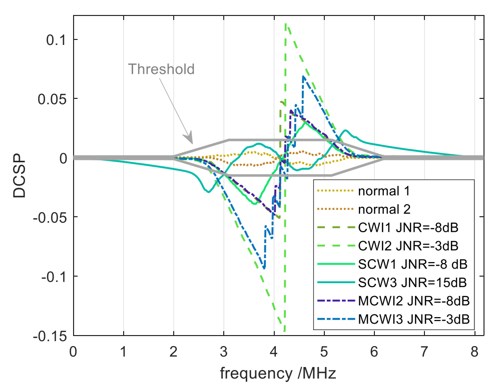


Fig. 13. Relationship of $\mathrm{DCSP}_{\mathrm{norm}}$ with threshold.

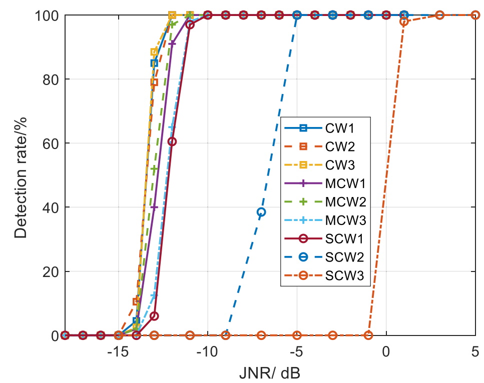
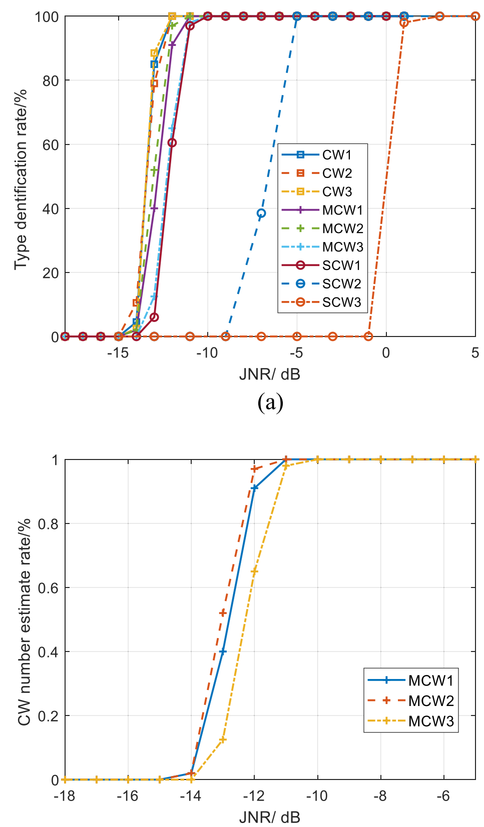


Fig. 14. Detection results of the jamming signals versus JNR.



(a)

(b)

Fig. 15. Identification results of the jamming signals versus JNR. (a) Identification rate. (b) Estimation rate of jamming components in MCWI.

the jamming signals are shown in Fig. 14. The detection results of various CWIs are similar and reach 100% in the situation of JNR above -12 dB, although it is approximately 0 when JNRs of CWIs are -15 dB. Under the MCWI environment, the number of CW components shows a slight influence on detection results. For MCWI with less than four components, the detection rate increases from 0% to 100% with JNR growing from -14 to -11 dB, while it increases to 100% until JNR $= -10$ dB in MCWI with five components. In SCWI cases, the detection ability is affected by jamming bandwidth. Specifically, for SCWI with B/2 bandwidth, the proposed method shows a high detection rate of 100% at JNR $= -10$ dB. Compared with narrow band SCW, the detection sensitivity broadband SCWI is lower. The broadband SCWI with the bandwidth of 1B and 1.5 B can be sensed as JNR over -9 dB and -1 dB, respectively. The detection results for broadband SCWI are acceptable as the influence of broadband SCWI with low JNR on GNSS is slight [34].

Once a jamming attack is detected, the jamming signals are further identified by features of DCSP. Fig. 15(a) presents the identification rate of the identification method proposed in Section IV. The identification rates are similar to the detection rates in Fig. 14, which demonstrates that the proposed method is able to correctly identify jamming types once the jamming attack is alarmed. Furthermore,

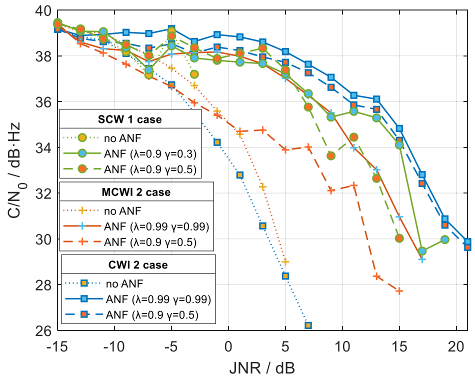


Fig. 16. C/N_0 in receivers with and without ANF.

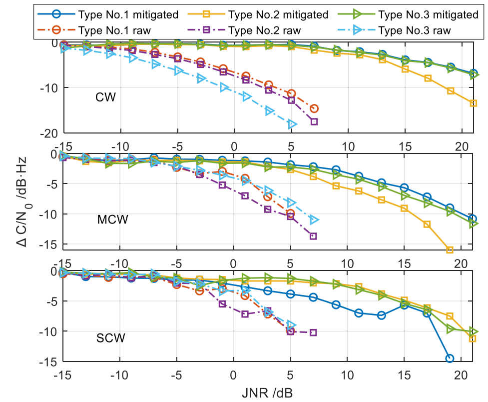


Fig. 17. C/N_0 degradation under different jamming cases.

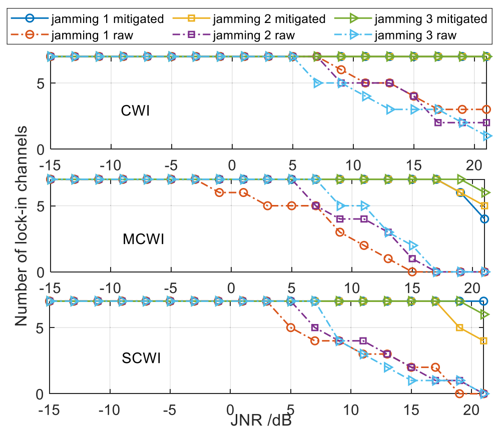


Fig. 18. Number of available satellites versus JNR.

as the number of jamming components in MCWI dictates the number of ANFs to be cascaded, Fig. 15(b) presents the rate of correctly estimating the number of jamming components in MCWI. Consistent with the identification results, the correctly estimating rate increases with the rise in JNR. An accuracy of over 95% in component estimation can be attained at −11 dB of JNR.

The identification results are used to activate the mitigation unit and the mitigation performance is evaluated in the following section.

C. Performance of Mitigation Method Using ANF With Switchable Iteration Factors

Three algorithms are performed in this study: no ANF, the conventional ANF, and the improved ANF proposed in this study. The configuration of the iteration factors for the conventional ANF is a fixed value, i.e., 0.9 of λ and 0.5 of γ, while the factors of the improved ANF are updated based on the identified jamming type. For the jamming signals with small frequency variations, such as CWI and MCWI, $\lambda = \gamma = 0.99$. For signals with obvious frequency variations, e.g., SCWI, $\lambda = 0.9$ and $\gamma = 0.3$.

The C/N_0 resulting from the three methods is compared. As is shown in Fig. 16, when ANF is not applied, the C/N_0 degrades significantly and channel losses lock once JNR climbs to 7 dB of CWI, 5 dB of MCWI and −3 dB of SCWI. The receiver with the conventional ANF effectively slows down the descend of C/N_0. Furthermore, the improved ANF with switchable iteration factors shows better performance than the conventional ANF. The C/N_0 of the improved ANF is 0.2–0.7 dB·Hz higher than the item of the conventional ANF in CWI case, and 4.6 dB·Hz and 4.1 dB·Hz higher in MCWI case and SCWI case.

It can be concluded that a higher C/N_0 can be achieved by the improved ANF with respect to the conventional ANF. Fig. 17 compares the C/N_0 degradation in a normal receiver and receiver with the proposed method. For the raw data in the normal receiver, the C/N_0 under all jamming attacks declines rapidly with the increment of JNR and tracking channel losses lock from JNR = 5 dB. When using the proposed method, the resulting C/N_0 is improved remarkably. The receiver is capable of lock-in even when JNR rises to 21 dB in CWI cases and 19 dB in MCWI and SCWI cases. As JNR rises, the C/N_0 improvement climbs correspondingly, which is up to 9–19 dB·Hz in different jamming cases.

The number of available channels is also a crucial performance indicator since at least four satellites are required to solve positions. Fig. 18 compares the number of lock-in channels in the normal receiver and the receiver with the proposed antijamming method. In CWI cases, the number of tracked satellites in the normal receiver declines to less than 4 from JNR 11 dB of CWI 1, JNR > 15 dB of CWI 2 and 3 attacks. However, all channels of antijamming receiver are kept locked-in even though JNR rises to 21 dB. For MCWI, tracking of the normal receiver is completely interrupted at JNR > 9 dB for MCWI 1, and JNR > 10 dB for MCWI 2 and 3. In the antijamming receiver, the number of locked-in channels is not less than 4 even if JNR raises to 21 dB. Similar results can be obtained for SCWI cases. From JNR > 9 dB, the number of the tracked channels are less than 4 in the normal receivers, while that in antijamming receiver is always over 4 as JNR increases to 21 dB.

The above-mentioned tests verify that the proposed method is capable of mitigating CWI, MCWI, and SCWI. The improvement of C/N_0 is up to 19 dB·Hz, 12 dB·Hz, and 9 dB·Hz for CWI, MCWI, and SCWI, respectively, and no less than four channels are available even JNR > 21 dB.

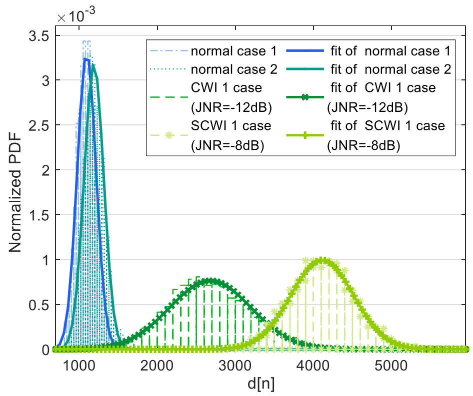


Fig. 19. Distribution of $d[n]$ under different cases.

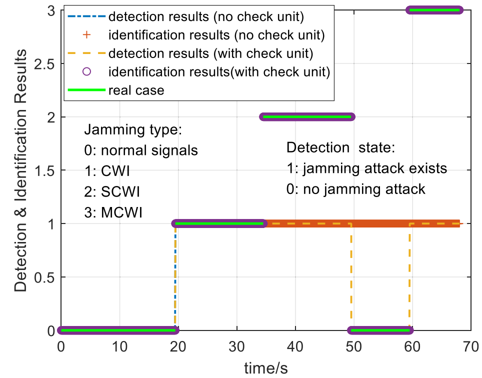


Fig. 20. Detection and identification of the changing jamming circumstance.

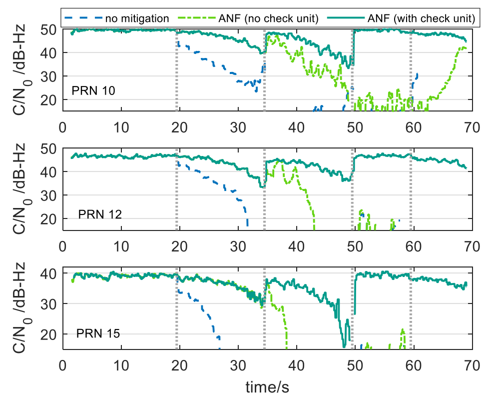


Fig. 21. C/N_0 results in the changing jamming circumstance.

D. Mitigation Performance Under the Varying Jamming Circumstance

In real circumstances, the jamming state can also be changed randomly. The check unit in Section V is designed to monitor the change of jamming attacks and reactivate the detection unit to update the ANF state.

The variable $d[n]$ in the ANF is adopted to monitor the variation of jamming. Fig. 19 shows the variable $d[n]$ under the normal case and the jamming environments. Consistent with the theoretical analysis, $d[n]$ reveals the energy of noise if filtered data of ANF is normal signals. It follows normal distribution with an expectation around 1200 and a variation around 14 400. Differently, $d[n]$ turns to the energy of noise and the jamming signals once jamming attack is filtered. Considering the jamming signals are too strong to be ignored, the expectation and variance of $d[n]$ are larger than that in the normal case. Therefore, $\lambda_{d,\mathrm{th}}$ is set as 1560. If $d[n]$ is over $\lambda_{d,\mathrm{th}}$, the filtered signals of the ANF are the jamming signals; otherwise, the mitigation of the current ANF is useless. The detection unit will be reactivated to reassess the received signals.

To assess the advancement of the check unit, a dynamic jamming case is designed. The jamming-free period is 0–19.5 s. Then, CWI 2 attack appears, and JNR grows from 0 dB with 3 dB step and 2 s dwell time. From 34.5 to 49.5 s, SCWI 1 attack takes the place of CWI and JNR increases in the same way. Following the second jamming-free period between 49.5 and 59.5 s, an MCWI 2 attack appears, exhibiting a similar JNR increment and persisting until 68 s.

Fig. 20 shows the detection and identification results of the antijamming method without and with a check unit, respectively. Although both methods can detect CWI attack initially, the method without the check unit fails to monitor the change of the jamming circumstance from 34.5 s. For the method with a check unit, the ANF with CWI matched factors cannot track SCWI well, which results in $d[n]$ drops to a low value. The check unit instantly monitors the variation and reactivates the detection unit. Then SCWI attack is identified and iteration factors of ANF are adapted to SCWI matched mode. Similarly, both the disappearance of jamming attack at 49.5 s and the appearance of MCWI at 59.5 s are monitored correctly.

Fig. 21 presents the C/N_0 results in the changing jamming circumstance. Stable tracking is achieved by the receiver at 1.3 s, enabling C/N_0 estimation thereafter. Until 19.5 s, no jamming attack exists and the C/N_0 keeps at high values. At 19.5 s, the CWI attack occurs, causing a sudden drop in C/N_0 of the normal receiver. Tracking loops lose lock at 28 s and cannot recover any more. The ANF can effectively filter CWI signals, enabling both mitigation methods to maintain lock-in for the tracking loops. When the jamming attack changes to SCWI at 34.5 s, the ANF without a check unit fails to recognize SCWI and tracking loops tend to lose lock from 39 s. However, the check unit observes the change of jamming state, and the proposed mitigation method updates and is still effective. Then, the C/N_0 returns to a normal value when jamming attack is absent from 49.5 s. At 60 s, the detection unit alarms the existence of a jamming attack. The identification unit identifies the jamming type as MCWI with four jamming components since four groups of local extrema are searched in the normalized DCSP. Then, four ANFs are cascaded to mitigate the impact of the MCWI attack.

The test results show that the check unit can monitor the change of jamming attacks. Once a jamming attack is changed, the detection unit is activated to update the ANF to counter different jamming circumstances adaptively.

VII. CONCLUSION

In this article, a lightweight and efficient antijamming method for GNSS single-antenna receivers is developed and it can detect, identify, mitigate, and monitor jamming attacks adaptively. Specifically, the jamming signals have been detected and identified by extracting the frequency characteristics of DCSP. The detection and identification sensitivity of over -13 dB JNR are achieved for CWI, MCWI, and narrowband SCWI, and those of over 0 dB are obtained for broadband SCWI. Furthermore, the ANF is modified through using switchable iteration factors, delivering a higher C/N_0. Comparing with the conventional ANF, the C/N_0 can be improved by 4.6 dB·Hz through introducing the switchable iteration factors in the modified ANF. More importantly, the state of jamming case is monitored immediately with the assistance of the proposed check unit in ANF. The receiver adopted the proposed method can cope with the common jamming signals with JNR $>$ 21 dB and the C/N_0 improvement can be up to 19 dB·Hz.

The antijamming method developed in this study is an independent software module before the signal processing stage. It can be used in receivers directly without changing other parts. Future studies will cover broadband noise and the integration of the proposed method with the advanced GNSS signals processing algorithms to further improve the performance of receivers.

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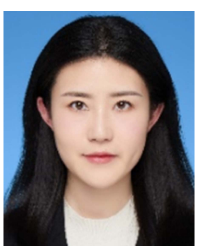

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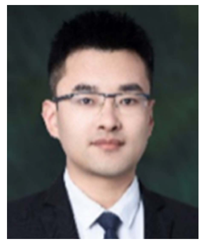

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